INCH-POUND

MIL-DTL-12560J (MR)
24 July 2009

SUPERSEDING
MIL-A-12560H (MR)
w/INT. AMENDMENT 4
20 July 2007
MIL-DTL-46177C (MR)
24 October 1998
(see 6.1.1)

DETAIL SPECIFICATION

ARMOR PLATE, STEEL, WROUGHT, HOMOGENEOUS (FOR USE IN COMBAT-VEHICLES AND FOR AMMUNITION TESTING)

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers wrought-steel homogeneous armor plate for ordered thicknesses from 0.098 inch (2.5mm) up to an ordered thickness of 6.000 inches (152.4mm), inclusive, for use in combat-vehicles and for ordered thicknesses from 0.500 inches (12.7mm) up to an ordered thickness of 12.000 inches (304.8mm), inclusive, for use in proof and acceptance testing of armor defeating ammunition (see 6.1, 6.2, and 6.3.3).

1.2 Classification. Wrought armor should be of the following classes as specified (see 6.2).

1.2.1 Class 1. Wrought armor plate with an ordered thickness of 0.098 to 6.000 inches (2.5mm to 152.4mm) which is heat treated to develop maximum resistance to penetration (see 6.1.1).

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army Research Laboratory, Weapons and Materials Research Directorate, Materials Applications Branch, Attn: RDRL-WMM-C, Aberdeen Proving Ground, MD 21005-5069 or emailed to rsquilla@arl.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil/.

AMSC N/A FSC 9515

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

1.2.2 Class 2. Wrought armor plate with an ordered thickness of 0.098 to 2.000 inches (2.5mm to 50.8mm) which is heat treated to develop maximum resistance to shock (see 6.1.2).

1.2.3 Class 3. Wrought armor plate with an ordered thickness of 0.625 to 12.000 inches (15.9mm to 304.8mm) which is heat treated to develop specific hardness and impact values for evaluation of armor defeating ammunition only. This class armor is not intended for use in combat-vehicles. To meet the thickness tolerances of this class armor special surface finishing may be required (see 6.1.3).

1.2.4 Class 4. Wrought armor plate with an ordered thickness of 0.098 to 2.000 inches (2.5 mm to 50.8 mm) is designed for either (a) better resistance to penetration with respect to class 1, or (b) better weldability with respect to class 1.

1.2.4.1 Class 4a. Wrought armor plate with increased hardness (see 3.2.6.1 and 6.1.4).

1.2.4.2 Class 4b. Wrought armor plate with reduced maximum carbon content (see 3.2.1 and 6.1.5).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129 - Military Marking for Shipment and Storage

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents are those cited in the solicitation or contract.

USADTC TOP 2-2-710 - Ballistic Tests of Armor Materials

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

ITOP 2-2-713	-	Ballistic Testing of Armor

(Application for copies is available online at http://www.dtic.mil/ or from the Defense Technical Information Center, 8725 John J. Kingman Road, Suite 0944, Fort Belvoir, VA 22060-6218.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF NONDESTRUCTIVE TESTING

ANSI/ASNT CP-189-2006	-	ASNT Standard for Qualification and Certification of Nondestructive Testing Personnel

(Copies of ANSI/ASNT publications are available from http:// www.asnt.org/shop/ or from the American Society for Nondestructive Testing, P.O. Box 28518, 1711 Arlingate Lane, Columbus, OH 43228-0518)

ASTM INTERNATIONAL

ASTM A6/A6M	-	Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes and Sheet Piling (DoD adopted)
ASTM A370	-	Standard Test Methods and Definitions for Mechanical Testing of Steel Products (DoD adopted)
ASTM A578/ A578M	-	Standard Specification for Straight-Beam Ultrasonic Examination of Rolled Steel Plates for Special Applications (DoD adopted)
ASTM A751	-	Standard Test Methods, Practices, and Terminology for Chemical Analysis of Steel Products (DoD adopted)
ASTM E10	-	Standard Test Method for Brinell Hardness of Metallic Materials (DoD adopted)
ASTM E18	-	Standard Test Methods for Rockwell Hardness of Metallic Materials (DoD adopted)
ASTM E23	-	Standard Test Methods for Notched Bar Impact Testing of Metallic Materials (DoD adopted)
ASTM E110	-	Standard Test Method for Indentation Hardness of Metallic Materials by Portable Hardness Testers (DoD adopted)
ASTM E290	-	Standard Test Methods for Bend Testing of Material for Ductility (DoD adopted)

(Copies of these documents are available from www.astm.org or ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order (see 6.2, 6.5 and 6.9), a sample or samples of the specified item shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.4. The contractor shall comply with this requirement at the time of his first order or contract and at any time that the supplier has not furnished the same class of wrought armor plate in the applicable thickness range under this specification within a period of 37 months. The approval of the first article samples authorizes the commencement of shipment but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification, namely conformance or production acceptance. The first article samples and test plates shall be manufactured by the process proposed for use on production armor. The manufacturer's declared chemical analysis shall be submitted to the contracting agency and to the ballistic test agency. The ballistic test agency shall record the first article ballistic test plates submitted, showing the dates tested. Requests from the procuring activity to the ballistic test agency as to prior conformance with first article tests shall be accompanied by copies of the first article test firing records. Any deviation(s) noticed by the ballistic agency shall be brought to the attention of the contracting activity and to the manufacturer.

3.1.1 First time producer. First time producers wishing to qualify to this specification shall follow the instructions of 6.7.

3.2 Production acceptance.

3.2.1 Chemical composition. A declared chemistry shall be submitted to the contracting agency or its authorized representative and to the ballistic test agency. The chemical composition of the declared chemistry shall conform to the requirements of Table I, column A, unless otherwise specified in the contract or purchase order (see 6.2). The chemical composition shall be determined by a product analysis (values shall be listed as weight percent) in accordance with 4.6.1.1 and 4.8.1. The first article samples and the production test plates shall utilize the same declared chemistry within the allowable ranges proposed for use in production. A statement showing the heat analysis of each melt and one product analysis of each lot and complete details of the heat treatment of each lot shall be furnished for the files of the purchaser at no cost. All elements of the chemical composition specified in Table I shall be shown in the statement. This statement shall be attached to the completed Rolled Homogeneous Armor Test Data Form (see Figure 1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

TABLE I. Chemical composition and precision (product analysis) [6/].

ELEMENT	COLUMN A MAXIMUM LIMIT for DECLARED CHEMISTRY (WEIGHT PERCENT)	COLUMN B [5/] ALLOWABLE RANGE for FUTURE PRODUCTION LOTS (WEIGHT PERCENT)
Carbon Class 1, 2, 3 & 4a	0.30 up to 2" thick, incl 0.33 over 2" up to 4" 0.35 over 4"	± 0.05
Carbon Class 4b	0.18 up to 1" thick, incl 0.22 over 1" up to 2", incl.	± 0.05
Manganese	NONE REQUIRED, HOWEVER IF: ≤ 1.00 > 1.00	 ± 0.15 ± 0.20
Phosphorus	0.020 [1/]	[4/]
Sulfur	0.010 [1/]	[4/]
Silicon	NONE REQUIRED, HOWEVER IF: ≤ 0.60 > 0.60 to ≤ 1.00 > 1.00	 ± 0.10 ± 0.15 ± 0.20
Nickel	NONE REQUIRED [3/]	± 0.25
Chromium	NONE REQUIRED, HOWEVER IF: ≤ 1.25 [3/] > 1.25	 ± 0.15 ± 0.25
Molybdenum	NONE REQUIRED, HOWEVER IF: ≤ 0.20 [3/] > 0.20	 ± 0.035 ± 0.075
Vanadium	NONE REQUIRED [3/]	± 0.05
Niobium	NONE REQUIRED [3/]	± 0.05
Boron	--- [2/]	[4/]
Copper	0.25 [3/]	[4/]
Nitrogen	0.03 [3/]	[4/]
Titanium	0.10 [3/]	[4/]
Zirconium	0.10 [3/]	[4/]
Aluminum	0.10 [3/]	[4/]
Lead	0.10 [3/]	[4/]
Tin	0.02 [3/]	[4/]
Antimony	0.02 [3/]	[4/]
Arsenic	0.02 [3/]	[4/]

[1/] Phosphorus and sulfur should be controlled to the lowest attainable levels.

[2/] When the amount of Boron is specified in the alloy, its content as determined by heat analysis shall not exceed 0.003 percent.

[3/] When the amount of an element is less than 0.02 percent the analysis may be reported as [< 0.02 wt%].

[4/] There are no limits on the allowable values for future lots; however, the values may not exceed those listed as the maximum limit.

[5/] Values are actual tolerance limits NOT percent tolerances.

[6/] Elements not listed in Table, but intentionally added, shall be reported.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

3.2.2 Carbon equivalence. Carbon equivalence (CE) shall be calculated for each heat per ASTM A6/A6M, i.e.

$$CE = C + [Mn/6] + [(Cr + Mo + V)/5] + [(Ni + Cu)/15]$$

where the elements are expressed in wt%.

3.2.2.1 Class 1 & 2. Class 1 and 2 steels shall have a CE of less than 0.80 wt% for plates less than 4 inches in thickness. For plates 4 inches and greater in thickness shall have a CE of 0.90 wt% or less.

3.2.2.2 Class 3. Class 3 steels shall have no CE limit unless otherwise specified in the contract or purchase order (see 6.2).

3.2.2.3 Class 4a. Class 4a steels shall have a CE of less than 0.80 wt% for plates less than 4 inches in thickness. For plates 4 inches and greater in thickness shall have a CE of 0.90 wt% or less.

3.2.2.4 Class 4b. Class 4b steels shall have a CE of less than 0.60 wt%.

3.2.3 Materials.

3.2.3.1 Structure. Processes of manufacture shall be such as to produce steel plate having, as nearly as practicable, a homogeneous structure throughout.

3.2.4 Heat treatment. All plates in each lot, including samples, shall receive the same heat treatment except for such variations in tempering temperature as shall be necessary to produce the prescribed hardness. The austenitizing temperature for production plates may vary within a range of 50° above the temperature used for test plates, but in no case shall it exceed 1700°F (see 6.6). The minimum tempering temperature for the various classes of armor are listed in Table II.

TABLE II. Minimum tempering temperatures.

CLASS	Minimum Tempering Temperature
Class 1	800 °F
Class 2	900 °F
Class 3	800 °F
Class 4	350 °F

3.2.5 Processing controls.

3.2.5.1 Heating. Stress relieving the plates, local or general, shall be allowed after final quenching and tempering at a maximum temperature of 50°F below the tempering temperature. Plates shall not be stress relieved in a temperature range of 600-800 °F. After stress relieving, the plates shall be tested for hardness and shall meet the requirements of Table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

REQUEST FOR BALLISTIC TEST OF ROLLED HOMOGENEOUS ARMOR			
SPECIFICATION: MIL-DTL-12560		**REVISION: J**	**AMENDMENT:**
FIRING RECORD:	**DATE:**	**TEST CONTRACT NO:**	

	PLATE/COIL MELTED	**PLATE/COIL ROLLED**	**PLATE/COIL HEAT TREATED**
Name:			
Address:			
POC:			
Phone No:			
Fax No:			

TEST ITEM IDENTIFICATION:		Ordered Thickness: _____
Lot No.__ _ _____ Plate No. _____	RHA Classification CLASS: ___ _ ___ Carbon Equivalence (CE):__ _____	Tempering Temp. _____ Austenizing Temp:__ _____
Bend Test:	Plate Thickness [Pass / Fail] 1)__ _ _____ 2)__ _ _____	Inside Radius 1)__ _ _____ 2)__ _ _____

PURPOSE: ___ Acceptance _ _ First Article ___ Reduced Testing

SAMPLE: _ __ Primary ___ Retest (Firing Record No. of Failed Sample_____)

CHEMISTRY ANALYSIS	C	Mn	P	S	Si	Ni	Cr	Mo
FIRST ARTICLE RESULTS:								
PRODUCTION RESULTS:								
ACCEPTANCE (Pass / Fail)								

V:	B:	N:	Zr:	Pb:	Sb:
Nb:	Cu:	Ti:	Al:	Sn:	As:

MECHANICAL PROPERTIES: **CHARPY IMPACT @ -40 Degs. F.** **Specimen Size:**

HARDNESS:	**LT Direction:1:_____ 2:_____ _ 3:_____**	**TL Direction:1:_ _____2:_____3:___ _**

BALLISTIC TEST RESULTS: **SEPARATELY HEAT TREATED TEST PLATE : YES____ _NO_____**

Test:	**Obl. (deg)**	**Actual Thickness (in)**	**Required V_{50} (fps)**	**Actual V_{50} (fps)**	**Pass/ Fail**
Projectile:					
	NOTES:				

Lot [met] [failed to meet] the ballistic requirements of specification MIL-DTL-12560

Government Representative:	Date:	Supplier Representative:	Date:

FIGURE 1. Rolled Homogeneous Armor (RHA) Test Data Form

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

3.2.5.2 Repairing. Weld repair is acceptable as long as it is done before the final heat treatment.

3.2.5.3 Forming. Forming after the final quenching and tempering operations may be performed if dye penetrant or magnetic particle inspection of the formed area finds it to be free of linear indications. The supplier shall practice such necessary process controls to prevent this condition.

3.2.5.4 Grinding. Grinding is not required but may be performed to remove minor defects not affecting the serviceability of the material provided that, unless otherwise authorized by the procuring activity, the grinding does not result in the reduction of the plate thickness in the ground area to less than the tolerances specified in ASTM A6/A6M or Table VI, as applicable.

3.2.6 Mechanical properties.

3.2.6.1 Surface hardness. The average surface hardness of each plate including first article samples shall be within the Brinell hardness range shown in Table III for the applicable class and thickness when tested in accordance with 4.8.2. The diameters of Brinell hardness impressions on any individual plate shall not vary by more than 0.15 mm.

TABLE III. Hardness requirements.

Specified nominal thickness of plate, inches		Armor class	Brinell hardness range, BHN, 3000 kg load	Brinell indentation diameters, mm	Rockwell C hardness range, HRC
From	To and incl.				
0.098	2.000	4a	420 - 460	2.98 - 2.85	45.0 - 48.3
0.098	0.624	1 & 4b	363 - 400	3.20 - 3.05	39.3 - 43.0
0.625	1.249	1, 3 & 4b	331 - 375	3.35 - 3.15	35.7 - 40.5
1.250	2.000	1, 3 & 4b	302 - 341	3.50 - 3.30	32.0 - 36.8
2.001	4.000	1 & 3	277 - 311	3.65 - 3.45	28.6 - 33.2
4.001	6.000	1 & 3	255 - 285	3.80 - 3.60	25.2 - 29.7
6.001	9.000	3	235 - 262	3.95 - 3.75	21.8 - 26.3
9.001	12.00	3	223 - 248	4.05 - 3.85	19.7 - 24.1
0.098	2.000	2	262 - 302	3.75 - 3.50	26.3 - 32.0

3.2.6.2 Impact. The Charpy V-notch impact test results of armor plate submitted shall meet the requirements shown in Table IV for the applicable hardness and thickness when tested in accordance with 4.8.3. The Charpy V-notch impact specimens shall be obtained in the TL orientation (transverse to the major direction of rolling with the notch perpendicular to the plate surface so that the crack propagates in the longitudinal direction).

3.2.6.3 Bend test. Bend test samples shall be required for a given lot of plates with an ordered thickness of 0.098 to 0.500 inches, inclusive and shall be capable of being bent to the requirements specified in Table V without cracking. For ordered thickness greater than 0.500 inches a bend test shall be as specified in the contract or purchase order (see 6.2). Two bend

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

test samples shall be tested in the transverse direction per ASTM E290, at room temperature through an included angle of 90 degrees (unrestrained) to the inside radii shown below. After bending, samples shall be free of cracks as determined visually.

TABLE IV. Minimum Charpy V-notch impact requirements (at -40°F + 2°F)

Average Brinell hardness, BHN	Minimum impact values (average of three(3) or more test specimens), ft-lb			
	Specimen Size			
	Standard	¾ width	½ width	¼ width
223	74			
229	71			
235	68			
241	65			
248	61			
255	58			
262	55			
269	51			
277	47			
285	43			
293	39			
302	35			
311	31			
321	28			
331	25	19		
341	22	17		
352	19	15		
363	17	13	9	4
375 - 460	16	12	8	4

TABLE V. Bend test requirements.

Ordered Plate Thickness (T)	**Inside Radius**	
	Class 2	Class 1, 3 & 4
0.098 to 0.500 inches, inclusive	3T	4T
Greater than 0.500 inches	as specified in contract or purchase order (see 6.2)	

3.2.7 Ballistic requirements. Ballistic requirements shall be in accordance with the Appendix A of this specification. When a complete penetration can not be obtained, the following rule shall be in effect until a new ballistic acceptance round can be utilized. When four (4) partial penetrations are above the minimum requirement for the specific thickness, the material shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

be certified as acceptable with a V_{50} (which obviously can not be specified) above the minimum requirement. There are no ballistic test requirements for class 2 plate thicker than 1.000 inch.

3.2.7.1 Ballistic test plate information. For each lot of armor plate a properly completed Rolled Homogeneous Armor (RHA) Test Data Form (see Figure 1) shall be submitted with each ballistic test plate that represents that particular processing lot.

3.2.8 Dimensions and tolerances.

3.2.8.1 Dimensions. Classes 1, 2, and 4 armor plates shall comply with the dimensions specified in the applicable drawings or in the contract or purchase order (see 6.2). Unless otherwise specified in the contract or order (see 6.2), dimensional tolerances will be in accordance with one-half the tolerances listed in ASTM A6/A6M. Class 3 armor tolerances on length and width dimensions shall be plus 1-1/2 inches and minus zero.

3.2.8.2 Thickness.

3.2.8.2.1 Thickness measurements (class 3 armor). The thickness of all class 3 plates shall be determined by measurements made one foot on centers over the entire surface of each plate. The measurements shall be to the nearest thousandth of an inch. The acceptance of each plate shall be based on these measurements meeting the thickness requirements of Table VI for class 3 armor.

TABLE VI. Thickness tolerances for class 3 armor plate. 1/

Specified plate thickness, inches		Permissible variation plus or minus, inches
Over	To and including	
0.250	0.500	0.015
0.500	1.125	0.020
1.125	1.499	0.025
1.499	1.749	0.030
1.749	2.000	0.035
2.000	3.000	0.040
3.000	4.000	0.045
4.000	6.000	0.075
6.000	8.000	0.083
8.000	10.000	0.098
10.000	12.000	0.122

1/ This armor is not intended for use in combat vehicles. To obtain these thickness tolerances special surfacing finishing shall be required.

3.2.8.2.2 Thickness tolerances. The thickness tolerances of each plate, after final treatment, shall be in accordance with one-half the tolerances listed in ASTM A6/A6M or with Table VI for Class 3 armor. The contract or purchase order (see 6.2) shall specify the thickness tolerance for Class 1 and Class 2 plate greater than 168 inches in width.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

3.2.8.3 Flatness. Unless otherwise specified in the contract or order (see 6.2), the flatness tolerance of each plate shall be within one-half of the tolerance listed in ASTM A6/A6M. Tighter tolerance requirements shall be specified in the contract or purchase order (see 6.2) and shall be as agreed upon between the contractor and the procuring activity.

3.2.8.4 Waviness. Unless otherwise specified in the contract or purchase order (see 6.2), the waviness tolerance of each plate shall be within one-half of the tolerance listed in ASTM A6/A6M.

3.2.9 Identification marking. Identification marking and records shall be such as to ensure positive identification of all plates, including test samples and specimens, with the lot and corresponding heat from which they were produced. First article and acceptance ballistic test plates shall also be marked with the manufacture's name or trademark, the number of this specification, and the ordered plate thickness in inches. First article plates shall be marked "PRE," acceptance plates "ACC," and retest plates shall be marked "R1" and "R2." If a second set of retest plates are submitted they shall be marked "RR1" and "RR2." The marking shall be approximately 1/16 inch deep and approximately 1/2 inch high and located along the edge with the lifting hole, if present, and shall be stamped into the surface unless the plate hardness makes painting necessary. All plate markings shall be highlighted with a highly visible yellow or orange border such as fluorescent paint or highlighted with a special white paint intended to delineate and highlight stamping. The primary plate rolling direction shall be identified. Manufacturers supplying armor, to be used for proving ground testing of ammunition, are required to:

a. Mark each class 3 plate with one foot square blocks showing thickness at the center of each block. The acceptance of each plate shall be based on these measurements meeting the thickness requirements set forth in Table VI.

b. Stamp each plate "HOMO, Class 3, MIL-DTL-12560, plate No. ____, heat No.____." Stamping shall be approximately 1/2 inch high letters to a depth approximately 1/16 inch. Location of stamping for all plates shall be in the upper right hand corner.

3.2.10 Information required. A statement showing the product analysis of each melt and complete details of the heat treatment of each lot shall be furnished for the files of the procuring activity in addition to the completed Rolled Homogeneous Armor (RHA) Test Data Form (see Figure 1). All elements of the chemical composition shall be shown in the statement, including special additives or hardening agents, whether shown in Table I or not.

3.2.11 Workmanship.

3.2.11.1 Surface imperfections. The top and bottom surface of each plate shall be free from the following surface defects: slivers, laps, checks, seams, blisters, snakes, cold shuts, cracks, burning, and laminations (see 6.3). The surface of each plate shall be such that mill scale or oxidation product shall not interfere with determination of acceptability. Imperfections listed above which are of such nature as to affect the fabrication of the material, are rejectable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

3.2.11.1.1 Depth of imperfections. The depth of rolled-in scale, scale pitting, mechanical gouges, or snakes shall not exceed 0.015 inch and shall not reduce the steel thickness below the allowable minimum. Isolated individual pits over 0.015 inch deep but not over 0.03 inch deep and not within 6 inches of each other and which do not violate the minimum allowable thickness, as specified in the applicable drawings and fabrication documents, are acceptable.

3.2.11.2 Edge preparation. Thermal cutting shall be permitted after final heat treatment provided the procedure, which may include grinding after thermal cutting, is such that no cracks develop on any thermally cut edge whether detected by nondestructive inspection, or as agreed upon in the contract. The heat affected zone of thermally cut plates (up to and including 1/2 inch in thickness) shall not exceed 1.2 times the plate's thickness from the cut edge. For plates over 1/2 inch thick, the heat affected zone shall not exceed 5/8 inch from the cut edge. In order to have the heat affected zone exceed these limits approval shall be obtained from the procuring activity and shall be as specified in the contract or purchase order (see 6.2).

3.2.11.3 Edge condition. Plate edge on plates delivered after heat treatment shall be free of cracks. The supplier shall practice such necessary process controls to prevent this condition.

3.2.11.4 Internal soundness. One plate from every lot that has an ordered thickness greater than 0.500 inch shall be ultrasonically examined for internal soundness in accordance with 4.7.3. The acceptance level shall be Level C as defined in ASTM A578/A578M, unless otherwise specified in the contract or purchase order (see 6.2).

3.2.11.4.1 Acceptance criteria.

3.2.11.4.1.1 Single linear indications. In any four inches of length a single linear indication shall not exceed twice the plate thickness.

3.2.11.4.1.2 Multiple linear indications. Multiple linear indications shall not exceed 1-1/2 times the plate thickness if two or more lie in the same plane. The total length of indications in one plane, in any four inch length, shall not exceed twice the plate thickness. No more than ten indications, whether in one plane or multiple planes, are permitted in any four inch length.

3.2.11.4.1.3 Cracks. All cracks shall be rejectable.

3.2.11.4.1.4 Removal of large indications. Large indications shall be removed by the manufacturer or processor by grinding, provided the resulting cavity does not exceed 1/4 inch.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Production acceptance inspection (see 4.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

4.2 Testing responsibility and facilities. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all the requirements as specified herein. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor may use his own or any other facilities suitable for the performance of the requirements specified herein, except ballistic tests (see 4.2.1), unless disapproved by the Government. The Government reserves the right to perform or check any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements and to determine the validity of the certifications.

4.2.1 Ballistic testing facility. Unless otherwise specified in the contract or purchase order (see 6.2), the ballistic test plates shall be forwarded to the Commander, USA ATC, ATTN: CSTE-DTC-AT-SL-V, Building 358, 400 Colleran Road, APG, MD 21005-5059 or to an approved Government facility for ballistic testing (see 6.8) for first article or production acceptance. Alternate ballistic testing facilities shall be considered as indicated in 6.8.

4.3 Lot. A lot shall consist of all production and ballistic test plates of the same melt of steel, of the same thickness, having the same treatment, and heat treated with the same thermal cycle in the same production furnace(s) in the same facility. When specified by the procuring activity (see 4.3.1 and 6.2), production and ballistic test plates shall be allowed to be heat treated separately. The test plate shall be heat treated in a production furnace.

4.3.1 Separately heat treated ballistic test plate. When the procuring activity allows a ballistic test plate to be heat treated separately from the production plates it represents (see 4.3), it shall be so stated in the data (see 6.6).

4.4 First article inspection. When required (see 6.2), the first article samples submitted in accordance with 3.1 shall be examined for all the provisions of this specification applicable to end item examination and shall utilize the same requirements and test methods as the production acceptance inspection shown in 4.5.

4.4.1 First article tests. First article tests shall consist of all the tests specified in 4.8.

4.4.2 First article ballistic test. Unless otherwise specified (see 3.1, 6.2, and 6.5), the first article ballistic test shall not be required provided (a) the manufacturer, within 37 months, has produced acceptable plates within the same nominal thickness ranges of Table VII, and (b) the production conditions are the same as for previously accepted plates. A supplier who has previously met the first article requirements shall furnish the procuring activity the firing record of the ballistic test plate and other pertinent data relative to compliance with first article. Neither first article nor acceptance ballistic tests shall be required for plates ordered to thicknesses of more than 6.25 inches. Acceptance of these thicker plates shall be based on meeting the other requirements of the specification.

4.5 Production acceptance inspection. The acceptance examination under 4.7 and the tests under 4.8 shall serve as a basis for the acceptance of individual production lots.

4.6 Sampling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

4.6.1 First article sampling.

4.6.1.1 Chemical composition. One sample for chemical analysis shall be taken from each plate selected for the ballistic tests.

4.6.1.2 Impact samples. At least one impact test sample shall be taken from each submitted test plate in accordance with 4.8.3.1 or 4.8.3.2 and each shall consist of enough material for three impact specimens.

4.6.1.3 Ultrasonic examination. Each submitted test plate 0.500 inch in thickness and thicker, shall be ultrasonically examined in accordance with 4.7.3.

4.6.1.4 Ballistic test plates. Two ballistic test plates for each nominal thickness range (see Table VII) shall be randomly selected and submitted for ballistic test. Any thickness within the range can be chosen to represent any other thickness in the range but the three plates submitted shall be of the same ordered thickness. The average thickness of each test plate shall be determined at the test site by the test agency as indicated in 4.6.2.5.

TABLE VII. Ballistic test plate sizes and corresponding test projectiles for first article and acceptance testing.

Nominal thickness range, inches	Armor Class	Minimum size [1/] of test plates, inches	Test projectile and obliquity	TABLE
0.098 to 0.250 incl.	1 & 4	12 x 36	Cal .30 M2 Ball at 30°	A-I
0.098 to 0.250 incl.	2	12 x 36	Cal .30 FSP at 0°	A-II
0.251 to 0.624 incl.	1 & 4	12 x 36	Cal .30 M2 AP at 0°	A-III
0.251 to 0.500 incl.	2	12 x 36	Cal .50 FSP at 0°	A-IV
0.625 to 1.125 incl.	1, 3 & 4	12 x 36	Cal .50 M2 AP at 0°	A-V
0.501 to 1.000 incl.	2	12 x 36	20 mm FSP at 0°	A-VI
1.126 to 2.000 incl.	1, 3 & 4	12 x 36	20 mm M602 APIT at 0°	A-VII
2.001 to 2.750 incl.	1 & 3	12 x 36	20 mm M602 APIT at 0°	A-VIII
2.751 to 3.500 incl.	1 & 3	60 x 60	90 mm M82 APC at 45°	A-IX
3.501 to 3.999 incl.	1 & 3	60 x 60	90 mm M82 APC at 30°	A-X
4.000 to 6.000 incl.	1 & 3	60 x 60	90 mm M318A1 AP at 0°	A-XI
Over 6.001 [2/]	3	---	---	---

[1/] Other sizes may be submitted for the convenience of the manufacturer if approved by the Armor Vulnerability Division, USACSTA, Aberdeen Proving Ground.

[2/] No ballistic test required.

4.6.2 Sampling for production acceptance inspection.

4.6.2.1 For chemical analysis. At least one sample for chemical analysis shall be taken from each heat in accordance with the applicable method specified in ASTM A751 (see 6.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

4.6.2.2 For hardness test. The Brinell hardness of each plate as heat treated in each lot up to 1-1/2 inches thick shall be measured in two places, one at each end of a diagonal on both plate surfaces. The Brinell hardness of each plate as heat treated in each lot over 1-1/2 inches thick shall be measured in four places, one at each end of a diagonal. Image analysis systems may be used to read the indent and compute the hardness and if the heat treating process is continuous (not batch or oscillation), then the amount of testing may be reduced to one measurement per plate.

4.6.2.3 For Charpy V-notch impact tests. At least one sample shall be taken from a plate representing each lot for Charpy V-notch impact tests. The sample shall be the same thickness as the plate they represent and large enough to obtain at least two specimens, in each direction, from each sample in accordance with 4.8.3.

4.6.2.4 Ultrasonic examination. All plates 1/2 inch and greater in thickness shall be examined ultrasonically. The ultrasonic examination shall be performed in accordance with 4.7.3.

4.6.2.5 For ballistic acceptance samples. Unless otherwise specified in the contract or order (see 6.2), one test plate shall be submitted from each lot. The minimum plate size shall be as shown in Table VII for the applicable thickness. The average thickness of each test plate shall be determined at the test site by taking the average of four thickness measurements. Measurements shall be made at least 1 inch from each edge but preferably at random in the intended impact area. Thickness measurements for plates up to and including 2.750 inch shall be read to the nearest 0.001 inch. Plates over 2.750 inch shall be read to the nearest 0.01 inch. If a producer submits plates outside the specified plate tolerances (see ASTM A6/A6M or Tables VI), the procuring activity shall be notified. A Rolled Homogeneous Armor (RHA) Test Data Form (see Figure 1) shall be completed and submitted with each ballistic test plate. In the event that plates of the same heat rolled to the same nominal thickness but with overlapping thickness tolerance ranges can be represented by one ballistic test plate, then only one of the ordered thicknesses need be submitted for acceptance testing. The other ordered thickness, however, shall be included on the applicable reporting form with the words indicating that it is represented by the sample to be tested. However, if the two ordered thicknesses are such that each thickness requires testing with a different type projectile as shown in Table VII, then each of the ordered thicknesses shall be ballistically tested.

4.7 Examination.

4.7.1 Visual. All steel plate shall be subject to visual inspection for compliance with the requirements for surface conditions (see 3.2.11 .1), edge quality (see 3.2.11.3 and 3.2.11.4) and identification marking (see 3.2.9).

4.7.2 Dimensions. All steel plates shall be subject to inspection for compliance with dimensional and tolerance requirements (3.2.8). Each class 3 armor plate shall divided into one foot squares painted on one surface and the results of each thickness measurement shall be painted in the square in which it was taken.

4.7.3 Ultrasonic examination.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

4.7.3.1 Inspection equipment. The ultrasonic soundness inspection equipment shall conform to ASTM A578/A578M.

4.7.3.2 Procedure. Unless otherwise specified (see 6.2), the ultrasonic examination shall be carried out in accordance with ASTM A578/A578M with the following exceptions.

(a) Scanning shall be continuous over 100% of the plate surface.
(b) Scanning rate shall be at a speed where recordable discontinuities can be detected.
(c) The testing frequency shall be a minimum of 2 megahertz (MHz).
(d) Any area within a plate where a discontinuity produces a continuous total loss of back reflection accompanied by continuous indications on the same plane that cannot be encompassed within a circle whose diameter is 1 inch shall be cause for rejection of that plate.

4.7.3.3 Certification of inspection personnel. Unless otherwise specified in the contract or purchase order (see 6.2), personnel performing ultrasonic inspection shall comply with the qualification requirements of ANSI/ASNT CP-189-2006. Personnel making accept/reject decisions in accordance with the process described by this specification shall be qualified to at least a level II.

4.8 Tests.

4.8.1 Chemical analysis. Chemical analysis shall be conducted in accordance with the applicable method specified in ASTM A751 (see 6.4). The analysis shall comply with the declared composition established in accordance with the requirement of Table I (see 3.2.1).

4.8.2 Hardness tests. Brinell hardness tests shall be conducted in accordance with either ASTM E10 or ASTM E110. Surface scale and decarburization shall be removed from the areas where the tests are to be made. However, no more than 0.060 inches shall be removed from the test area.

4.8.2.1 Hardness tests on impact test specimens. Samples for all classifications shall be examined on a lot by lot basis. Except as provided in 4.8.2.2, each sample from which impact test specimens are taken shall have a hardness check made upon it as follows: For samples 0.098 to 0.500 inch inclusive the hardness check shall be made on the surface. For samples of 1/2 inch up to 4 inches in thickness, at least two hardness tests shall be taken from an adjacent location, midway between the surfaces of the plate. For samples of 4 inches in thickness or greater, at least six hardness tests shall be made at equal intervals across the plate thickness.

4.8.2.2 Rockwell hardness tests on impact specimens. At the option of the contractor at least two Rockwell C hardness tests shall be made on each impact test specimen sampled from each lot. The test specimens shall be prepared and tested in accordance with ASTM E18. The average shall not exceed the maximum requirement for thickness specified in Table III.

4.8.2.3 Hardness traverse test. Prior to production to assure compliance to paragraph 3.2.11.2, the size of the heat affected zone for thermally cut edges shall be determined by taking Rockwell C hardness tests at 1/16 inch increments from the edge. Surface scale and

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

decarburization shall be removed from the test surface. The distance at which there is no change in surface hardness shall be considered the depth of the heat affected zone. The contractor shall document the test results for each thickness and thermal cutting method used in production. The contractor is also responsible for documenting all heat input controls to assure the cutting process is under control. Re-verification of heat affected zone size for subsequent production lots is not required provided there is no change in the cutting method or heat input from that previously tested. The records shall be available for review by the cognizant government representative.

4.8.3 Charpy V-notch impact tests. At least three Charpy V-notch impact test specimens shall be taken in the transverse (TL) direction from each sample as obtained in accordance with 4.8.3.1 or 4.8.3.2, and shall be prepared and tested in accordance with ASTM E23 and ASTM A370.

4.8.3.1 For samples less than 4 inches in thickness. Charpy V-notch impact test specimens from samples less than 4 inches in thickness shall be taken in the TL orientation (see 3.2.6.2) from a location midway between the top and bottom surfaces of the plate and at least 4 inches or 2T, whichever is less, from any quenched edge as well as outside the heat affected zone of any thermal-cut edge. When the amount of materials available is insufficient to obtain standard specimens, the largest attainable subsize Charpy V-notch impact specimens shown in Figure 4 of ASTM E23 shall be used and the results compared to the applicable requirements of Table IV. There are no Charpy V-notch impact test requirements for samples less than 0.140 inches thick.

4.8.3.2 For sample 4 inches or greater in thickness. Charpy V-notch impact test specimens from samples 4 inches or greater in thickness shall be approximately 1 inch below the surface of the plate and at least 4 inches from any quenched edge as well as outside the heat-affected zone of any thermally-cut edge.

4.8.4 Ballistic tests. Ballistic testing of armor plate shall be conducted at a Government approved test facility specified in 4.2.1, unless otherwise specified in the contract or purchase order (see 6.2). Testing shall be conducted in accordance with the requirements of the Appendix A.

4.9 Reduced testing. At the discretion of the procuring activity, the amount of testing shall be reduced provided the results on consecutive lots indicate that a satisfactory uniform product meeting the testing requirements is being produced (see 6.2). Reduced testing shall be in accordance with a system previously approved or established by the procuring activity involved.

4.10 Rejection and retest.

4.10.1 Rejection. Unless otherwise specified in the contract or order (see 6.2), failure of the first article samples to meet the requirements of this specification shall be cause for rejection of the process, and failure of the acceptance samples to meet the requirements of this specification shall be cause for rejection of the lot (see 4.10.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

4.10.2 Retest. Unless specific retest procedure is specified in the contract or order (see 6.2), two retest samples shall be submitted for each failed sample. Failure of either of the retest samples (plates) shall be cause for rejection of the material. First article retests shall not be permitted until the supplier has made the necessary correction in the processing of the material to the satisfaction of the procuring activity.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The armor specified herein is intended for use on combat vehicles and/or proof and acceptance testing of armor defeating ammunition, but should also be applicable to armor for emplacements, shields, pill boxes, testing, and the like.

6.1.1 Class 1. Class 1 wrought armor is intended for use in those areas where maximum resistance to penetration by armor piercing types of ammunition is required. Please note that MIL-DTL-46177, entitled, "ARMOR, STEEL PLATE AND SHEET, WROUGHT, HOMOGENEOUS (1/8 TO LESS THAN 1/4 INCH THICK)" was incorporated into this document for new design. MIL-DTL-46177C has been in-activated for new designs; therefore, legacy designs/items (contracts/drawings) which used MIL-DTL-46177C in the past can continue to use MIL-DTL-46177C. The current thickness of this specification was even further reduced to 0.098 inches.

6.1.2 Class 2. Class 2 wrought armor is intended for use in those areas where maximum resistance to failure under conditions of high rates of shock loading is required and where resistance to penetration by armor piercing ammunition is of secondary importance. It is intended for use as protection against anti-tank land mines, hand grenades, bursting shells, and other blast-producing weapons.

6.1.3 Class 3. Class 3 wrought armor is intended for use in the proof and acceptance testing of armor defeating ammunition. This class is not intended for use in combat vehicles. Lift hole requirements for Class 3 armor should be specified by the procuring activity.

6.1.4 Class 4a. Class 4a wrought armor has the same maximum carbon content as the other classes, but is tempered at lower temperatures (in Stage I) to give higher hardness. This higher hardness gives the armor enhanced resistance to ballistic penetration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

6.1.5 Class 4b. Class 4b wrought armor has a lower carbon content than the other classes, but is tempered at lower temperatures (in Stage I) to give the same mechanical and ballistic properties as Class 1 armor. This lower carbon content gives the armor enhanced weldability.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Specify ordered thickness (see 1.1 and 3.2.8.1) and classification (see 1.2).
c. If issues of documents are different (see 2.2.1).
d. If a different issue is to be used (see 2.2.2 and 2.3)
e. If first article samples are required (see 3.1, 4.4, and 4.4.2).
f. If the declared chemistry composition can be different (see 3.2.1).
g. A CE limit for Class 3 material, if required (see 3.2.2.2).
h. A bend test requirement if a bend test is required for material greater than 0.500 inch in thickness (see 3.2.6.3 and Table V).
i. If dimensional tolerances can be other than in accordance with one-half the tolerances listed in ASTM A6/A6M (see 3.2.8.1).
j. Specify thickness tolerances for Class 1 and Class 2 plate greater than 168 inches in width (see 3.2.8.2.2).
k. If the flatness tolerance is different than those specified (see 3.2.8.3).
l. If tighter flatness tolerance requirements are to be specified (see 3.2.8.3).
m. If the waviness tolerance is different than those specified (see 3.2.8.4).
n. If the limits specified for heat affected zone need to be exceeded (see 3.2.11.2).
o. Specify the acceptance level for internal soundness if other than Level C as defined in ASTM A578/A578M (see 3.2.11.4)
p. If someone other than the contractor is responsible for the performance of all the requirements of the specification (see 4.2).
q. If the contractor can't use his own facility or any other facility for testing (see 4.2).
r. If the ballistic tests are to be conducted at another location (see 4.2.1and 4.8.4).
s. If production and ballistic test plates can be heat treated separately (see 4.3).
t. If the number of test plates per lot for ballistic acceptance testing is different (see 4.6.2.5).
u. If ultrasonic examination is to be different (see 4.7.3.2).
v. If certification of inspection personnel is different (see 4.7.3.3).
w. Specify reduced testing plan when applicable (see 4.9).
x. If rejection requirements differ (see 4.10.1).
y. If retest requirements differ (see 4.10.2).
z. Packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Contractor. The contractor or prime contractor is the company which has a direct contract from the Government to furnish an end item, usually a vehicle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

6.3.2 Contracting officer. The term “contracting officer’ means the person executing a contract on behalf of the Government and any other officer or civilian employee who is properly designated contracting officer; and the term includes, except as otherwise provided, the authorized representative of a contracting officer acting within the limits of his authority.

6.3.3 Homogeneous rolled armor. Homogeneous rolled armor is armor having uniform composition heat treatment.

6.3.4 Manufacturer. The manufacturer is defined as the company producing the steel alloy plate.

6.3.5 Procuring activity. The term “procuring activity” is that activity of the Government which actually initiates the request for procurement and maintains the records of the procurement.

6.3.6 Slivers. An imperfection consisting of a very thin elongated piece of metal attached by only one end to the parent metal into whose surface it has been worked.

6.3.7 Laps. A surface imperfection with appearance of a seam caused by hot metal, fins or sharp corners being folded over and thus being forged or rolled into the surface but without being welded.

6.3.8 Checks. Numerous very fine cracks at the surface of a metal part. Checks may appear during processing or during service and are most often associated with thermal cycling or thermal treatment. Also called check marks, checking, and heat checks.

6.3.9 Seams. An unwelded fold or lap that appears as a crack, usually resulting from a discontinuity on a metal surface.

6.3.10 Blisters. A raised area, often dome shaped, resulting from delamination under pressure of expanding gas trapped in a metal in a near sub-surface zone. Very small blisters may be called pinhead blisters or pepper blisters.

6.3.11 Snakes. Any crooked surface imperfection in a metal plate, resembling a snake.

6.3.12 Cold shuts. Freezing of the top surface of an ingot before mold is full.

6.3.13 Burning. Permanently damaged metal due to overheating enough to cause incipient melting or intergranular oxidations. Note: this condition is usually obscured by normal cleaning methods and would require deep pickling and/or metallography to note the continuous oxidation (chicken wire effect) of the enlarged grain boundaries.

6.3.14 Lamination. A type of discontinuity with separation or weakness generally aligned parallel to direction of the worked surface of the metal and may be the result of pipe, blisters, seams, inclusions, or segregation; elongated and made directional by working.

6.3.15 Pit. A cavity or depressed area on the surface of a plate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

6.3.16 Linear indication. For nondestructive examination purposes, a linear indication is evidence of a discontinuity that requires interpretation to determine its significance.

6.3.17 Thickness. The symbol “T” is used throughout this specification to indicate the nominal thickness of the plate under consideration.

6.3.18 Crack. A crack is a planar discontinuity in the metal that has length and depth. Cracks may be surface cracks or through cracks.

6.4 Chemical analysis. Suggested ASTM instrumental methods that can be used for chemical analysis are E322 and E415. ASTM A751 should be consulted for a complete list of methods.

6.4.1 Changes in composition. Changes in composition may not necessarily be cause for retesting but should be subject to review by the procuring activity to determine whether additional first article testing is required.

6.5 Special first article ballistic test. Special first article ballistic tests are required when the manufacturer changes either the melting procedure or heat treatment or the declared chemistry of the armor.

6.6 Production plates. Material made to this specification has a tendency to develop stress cracks if not tempered as soon as possible after austenitizing treatment. To avoid this situation all plates should be left in the hot rolled or tempered condition while waiting for the ballistic test results.

6.7 Potential suppliers. Potential suppliers who have not previously supplied armor plate to MIL-A-12560 and wish to have their material ballistic tested may do so at their own expense. It is recommended that inquiries for such testing be directed to Commander, USA ATC, ATTN: CSTE-DTC-AT-SL-V, Building 358, 400 Colleran Road, APG, MD 21005-5059.

6.8 Alternate ballistic testing facility. Request for approval for an alternate ballistic testing facility should be forwarded by the procuring activity to the Director, U.S. Army Research Laboratory, Weapons and Materials Research Directorate, Specifications and Standards Office, Attn: RDRL-WMM-C, Aberdeen Proving Ground, MD 21005-5069 and should be obtained prior to the contract award.

6.9 New contracts sponsored by Government agencies. At the time that a new contract is initiated for the production of combat vehicles, the contractor’s supplier is to estimate the number, size and delivery schedule of the ballistic test plates which are to be submitted for first article or acceptance testing. A lead time of 60 days after the contract has been signed is to be allowed prior to shipment of the first ballistic test plate(s) to APG to insure that all administrative functions for the establishment of a new ATC project have been completed in preparation for the test. The contracting government agency is to initiate the new project through a letter to Commander, USA ATC, ATTN: CSTE-DTC-AT-SL-V, Building 358, 400 Colleran Road, APG, MD 21005-5059 requesting a cost estimate for the ballistic testing of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

applicable number of sizes of plates. In the case of increases in scope of existing projects, similar correspondence is needed.

6.10 Fabrication. The armor plate covered by this specification is subject to fabrication involving cutting, drilling, forming and welding. It is intended that selection and control of chemical composition, cleanliness, and plate processing should be such that the armor will be suitable for fabrication in accordance with TACOM code 12479550, TACOM Ground Combat Vehicle Welding Code-Steel. Copies of this document are available from U.S. Army Tank-Automotive and Armaments Command, Warren, MI 48397-6000.

6.11 Metric units. When metric dimensions are required, units for inch, foot, foot-pounds and feet per second may be converted to the metric equivalent by multiplying them by the following conversion factors listed in Table VIII.

TABLE VIII. Conversion factors.

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/sec	0.3048	=	meter per second (m/s)
Fahrenheit (°F)	(t°F - 32)/1.8 = t°C		Centigrade (°C)

Note: Conversion factors can be associated with ASTM E380 entitled “Metric Practice Guide.”

6.12 Subject term (key word) listing.

Ballistic limit
Ballistic plate
M2 projectile
M318 projectile
M602 projectile
M70 projectile
M82 projectile

6.13 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

APPENDIX A

BALLISTIC TESTING OF ARMOR, STEEL PLATE, WROUGHT, HOMOGENEOUS

A.1 SCOPE

A.1.1 This appendix covers the requirements for ballistic testing of wrought homogeneous steel armor plate.

A.2 DEFINITIONS

A.2.1 Fair impact.

A.2.1.1 Caliber .30 AP through 20 mm tests. A fair impact is an impact resulting from the striking of the test plate by a projectile in normal flight (no excessive yawing or tumbling) and separated from another impact or the edge of the plate, hole, crack or spalled area by an undisturbed area at least two test projectile diameters.

A.2.1.2 57 mm AP and larger calibers. A fair impact is an impact resulting from the striking of the test plate by a projectile in normal flight (no yawing or tumbling) and separated from another impact or from the edge of plate, hole, crack or spalled area by at least one test projectile diameter of undisturbed area.

A.2.2 Witness plate. A witness plate is normally a 0.014 inch thick sheet of 5052 aluminum alloy (or a 0.020 inch thick sheet of 2024-T3 aluminum alloy placed 6 inches (+ 1/2 inch) behind and parallel to the test plates or other ballistic sample.

A.2.3 Complete penetration, protection, CP (P).

A.2.3.1 For caliber .30 M2 AP through 20 mm M602 APIT. A protection complete penetration occurs when the projectile or one or more fragments of a projectile or plate pass beyond the back of the test plate and perforates the witness plate.

A.2.3.2 For projectiles larger than 20 mm. A protection complete penetration occurs when one or more fragments of a projectile or plate has been ejected from the rear of the plate as determined by visual inspection.

A.2.4 Partial penetration, protection, PP (PI). A partial penetration is any impact that is not a complete penetration.

A.2.5 Gap. The difference in velocity between the high partial penetration velocity and the low complete penetration velocity used in computing the ballistic limit where the high partial penetration velocity is lower than the low complete penetration velocity.

A.2.6 Protection ballistic limit, BL (P).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

A.2.6.1 Caliber .30 M2 AP caliber .50 M2 AP, 20 mm M602 APIT penetration tests. The BL (P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL (P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Firing shall continue until either a 4 round BL (P) having a maximum velocity spread of 60 ft/sec or a 6 round BL (P) having a maximum velocity spread of 90 ft/sec has been attained, whichever comes first in the normal sequence of firing. If both occur simultaneously, the 6 round BL (P) shall be reported.

A.2.6.1.1 Difference between the high partial penetration velocity and the low complete penetration. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the PP (P) velocity being higher than the low CP (P) velocity) exceeds 90 ft/sec, the firing data shall be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity is equal to or above the minimum specified ballistic limit velocity for the plate thickness, the ballistic limit shall be computed on the basis of 4 or 6 rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity then testing shall continue until a ten round ballistic limit has been attained using the smallest possible velocity spread.

A.2.6.2 57 mm M70 AP and larger caliber projectiles. For the purpose of this specification, the protection ballistic limit is the average of four fair protection criteria impact velocities comprising the first two lowest velocities in the firing order resulting in complete penetrations and the first two highest velocities in the firing order resulting in partial penetrations which meet the condition that the velocity spread for the 4 rounds shall not exceed 100 ft/sec.

A.2.6.3 Reduction of large velocity gap in borderline cases. If the ballistic limit which has been determined is within ± 10 ft/sec from the minimum allowable ballistic limit and a gap exists which is greater than 25 ft/sec, then another round, or rounds, shall be fired to reduce the gap to 25 ft/sec or less. The ballistic limit shall then be recomputed using the criteria of paragraphs A.2.6.1 or A.2.6.2 as applicable. The recomputed BL (P) shall be reported as the BL (P) of the plate. In borderline cases a reduction of the gap between the high partial penetration velocity and the low complete velocity shall result in a better evaluation of the BL (P).

A.3 REQUIREMENTS

A.3.1 Resistance to penetration. The minimum acceptable ballistic limits shall be in accordance with the values shown in Tables A-I through A-XI.

A.3.1.1 Class 3 armor plate. For Class 3 armor plate, upper ballistic limit (maximum) values shall be in accordance with the values shown in Tables A-V, A-VII through A-XI.

A.3.2 Back spalling.

A.3.2.1 Plates 1/2 to 11/16 inch in thickness. Plates of these ordered thicknesses shall be subjected to a back spall test consisting of firing two rounds of 20 mm M95 AP ammunition at a 0° obliquity

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

at the test plate, the striking velocity shall be in the velocity range of 2500-2550 ft/sec. The maximum allowable back spall dimension shall be 1 5/8 inches.

A.3.2.2 Plates 3/4 inch and greater in thickness. Spalling shall not exceed 2.0 test projectile diameters after proof firing at normal obliquity in accordance with the appropriate table of the appendix, and shall not exceed 2.5 test projectile diameters in the case of oblique penetration tests.

A.3.3 Cracking.

A.3.3.1 0° obliquity attack. Immediately after ballistic testing, plates subject to oblique attack shall not develop any crack which, when measured from the center of impact, extends outside a circle having a radius equivalent to twice the test projectile diameter.

A.3.3.2 Oblique attack. Immediately after ballistic testing, plates subjected to 0° obliquity attack shall not develop any crack which when measured from the center of impact, extends outside a circle having a radius equivalent to twice the test projectile diameters.

A.3.4 Rejection. Failure to meet any of the above requirements shall be cause for plate rejection.

A.4 TESTS

A.4.1 Ballistic tests. V_{50} ballistic tests shall be performed in accordance with USATECOM TOP 2-2-710, Ballistic Tests of Armor Materials to determine compliance with the requirements of Tables A-I through A-XI.

A.4.1.1 Plate thickness. Plate thickness as measured by the ballistic test agency shall be used to determine the required ballistic limit of the plate. The required ballistic limit shall be determined by interpolation of the tables in the appendix, if necessary.

A.4.1.2 Lift hole requirement for test plates.

A.4.1.2.1 Ballistic acceptance testing plates.

a. Ballistic acceptance test plates for either class 1 or armor which are less that 2.751 inches ordered thickness are to be supplied without a lifting hole.
b. Ballistic acceptance test plates for either class 1 or 3 armor which are 2.751 inches through 6.25 inches thick shall be provided with 3-inch diameter lifting holes, one in each of two adjacent sides as shown in Figure A-1.

A.4.1.3 Rejection and retest of ballistic plates.

A.4.1.3.1 First article tests (rejection). Unless noted otherwise in the contract or order, failure of any one or more of the three first article test plates to meet the minimum ballistic requirements as specified in the appendix to the specification indicates failure of the products and process.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

A.4.1.3.2 First article (retests). Resubmission of ballistic retest plates shall not be made until the manufacturer has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Three retest plates shall be submitted for first article testing and all three shall pass.

A.4.1.3.3 Acceptance tests (rejection). Unless otherwise noted in the contract or order, failure of a test plate to meet the ballistic requirement indicates failure of the lot; however, the final decision shall depend on the outcome of retests, if submitted.

A.4.1.3.4 Acceptance tests (retests). If a test plate representing a lot fails to meet the ballistic requirement, the manufacturer has the following options: Immediately upon notification of the failure:

(1) At manufacturer's expense submit two additional test plates from the same lot for ballistic retest, or
(2) First re-heat treat (quenching and tempering) the lot and then submit a test plate from the re-treated lot, or
(3) Scrap the lot and submit a plate representing a new lot for acceptance.

If the manufacturer chooses any one of these options and the ballistic retest plate (or plates) meet the requirements then the lot represented is acceptable. If he chooses option (1) and one or both the retest plates fail, the manufacturer shall re-heat treat the lot and submit a test plate from the re-treated lot. If this plate fails the lot is rejected. If the manufacturer chooses option (3) and the test plate fails, the manufacturer shall again resort to any one of the three options.

A.4.1.4 Disposition of ballistic test plates.

A.4.1.4.1 First article test plates. Upon request of the applicant within 15 days after ballistic testing, first article plates shall be returned "as is" to the applicant, at the applicant's expense, unless the plates were destroyed in testing.

A.4.1.4.2 Acceptance test plates. Acceptance test plates that comply with the requirements of this specification are considered as part of the lot of steel they represent and ownership of them passes to the Government with the acceptance of that lot. Acceptance test plates that fail to comply with the requirements of this specification are considered as part of the lot they represent and remain the property of the producer just as the rejectable lot does. The failed plates shall be returned, upon request, as in A.4.1.4.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-I. Minimum required V50 ballistic limits (protection criteria) for Class 1 & 4 armor, firing obliquity: 30°; projectile: caliber .30 ball M2.

Thickness, inches	Required BL(P), ft/sec **CLASS 1 & 4**	Thickness, inches	Required BL(P), ft/sec **CLASS 1 & 4**	Thickness, inches	Required BL(P), ft/sec **CLASS 1 & 4**
0.087	1455	0.120	1762	0.153	2071
0.088	1464	0.121	1771	0.154	2080
0.089	1473	0.122	1781	0.155	2090
0.090	1482	0.123	1790	0.156	2099
0.091	1491	0.124	1799	0.157	2109
0.092	1500	0.125	1809	0.158	2118
0.093	1509	0.126	1818	0.159	2127
0.094	1519	0.127	1828	0.160	2137
0.095	1528	0.128	1837	0.161	2141
0.096	1537	0.129	1846	0.162	2146
0.097	1546	0.130	1856	0.163	2151
0.098 1/	1556	0.131	1865	0.164	2156
0.099	1565	0.132	1874	0.165	2161
0.100	1574	0.133	1884	0.166	2166
0.101	1584	0.134	1893	0.167	2171
0.l02	1593	0.135	1903	0.168	2176
0.103	1603	0.136	1912	0.169	2180
0.104	1612	0.137	1921	0.170	2185
0.105	1622	0.138	1931	0.171	2190
0.106	1631	0.139	1940	0.172	2195
0.107	1640	0.140	1949	0.173	2200
0.108	1650	0.141	1959	0.174	2204
0.109	1659	0.142	1968	0.175	2209
0.110	1668	0.143	1977	0.176	2214
0.111	1678	0.144	1987	0.177	2219
0.112	1687	0.145	1996	0.178	2224
0.113	1696	0.146	2006	0.179	2228
0.114	1706	0.147	2015	0.180	2233
0.115	1715	0.148	2024	0.181	2238
0.116	1725	0.149	2034	0.182	2242
0.117	1734	0.150	2043	0.183	2247
0.118	1743	0.151	2052	0.184	2252
0.119	1753	0.152	2062	0.185	2257

1/ Specification requirements begin for this ordered thickness

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-I. Minimum required V50 ballistic limits (protection criteria) for Class 1 & 4 armor, firing obliquity: 30°; projectile: caliber .30 ball M2. (Continued)

Thickness, inches	Required BL(P), ft/sec **CLASS 1 & 4**	Thickness, inches	Required BL(P), ft/sec **CLASS 1 & 4**	Thickness, inches	Required BL(P), ft/sec **CLASS 1 & 4**
0.186	2261	0.215	2393	0.244	2518
0.187	2266	0.216	2398	0.245	2523
0.188	2271	0.217	2402	0.246	2527
0.189	2275	0.218	2407	0.247	2531
0.190	2280	0.219	2411	0.248	2535
0.191	2285	0.220	2416	0.249	2539
0.192	2289	0.221	2420	0.250 2/	2544
0.193	2294	0.222	2424	0.251	2548
0.194	2298	0.223	2428	0.252	2552
0.195	2303	0.224	2433	0.253	2556
0.196	2308	0.225	2437	0.254	2560
0.197	2312	0.226	2441	0.255	2564
0.198	2317	0.227	2446	0.256	2568
0.199	2321	0.228	2450	0.257	2573
0.200	2326	0.229	2454	0.258	2577
0.201	2330	0.230	2459	0.259	2581
0.202	2335	0.231	2463	0.260	2585
0.203	2339	0.232	2467	0.261	2589
0.204	2344	0.233	2472	0.262	2593
0.205	2348	0.234	2476	0.263	2597
0.206	2353	0.235	2480	0.264	2601
0.207	2357	0.236	2484	0.265	2605
0.208	2362	0.237	2489	0.266	2610
0.209	2366	0.238	2493	0.267	2614
0.210	2371	0.239	2497	0.268	2618
0.211	2375	0.240	2501	0.269	2622
0.212	2380	0.241	2506	0.270	2626
0.213	2384	0.242	2510	0.271	2630
0.214	2389	0.243	2514	0.272	2634

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-II. Minimum required ballistic limits (protection criteria) for Class 2 armor, firing obliquity: 0°, projectile: caliber .30 FSP

Thickness, inches	Required BL(P), ft/sec	Thickness, inches	Required BL(P), ft/sec	Thickness, inches	Required BL(P), ft/sec
0.087	1102	0.120	1489	0.153	1875
0.088	1114	0.121	1500	0.154	1887
0.089	1125	0.122	1512	0.155	1899
0.090	1137	0.123	1524	0.156	1911
0.091	1149	0.124	1536	0.157	1922
0.092	1160	0.125	1547	0.158	1934
0.093	1172	0.126	1559	0.159	1946
0.094	1184	0.127	1571	0.160	1958
0.095	1196	0.128	1582	0.161	1969
0.096	1207	0.129	1594	0.162	1981
0.097	1219	0.130	1606	0.163	1993
0.098 1/	1231	0.131	1618	0.164	2004
0.099	1242	0.132	1629	0.165	2016
0.100	1254	0.133	1641	0.166	2028
0.101	1266	0.134	1653	0.167	2040
0.l02	1278	0.135	1664	0.168	2051
0.103	1289	0.136	1676	0.169	2063
0.104	1301	0.137	1688	0.170	2075
0.105	1313	0.138	1700	0.171	2086
0.106	1325	0.139	1711	0.172	2098
0.107	1336	0.140	1723	0.173	2110
0.108	1348	0.141	1735	0.174	2122
0.109	1360	0.142	1747	0.175	2133
0.110	1371	0.143	1758	0.176	2145
0.111	1383	0.144	1770	0.177	2157
0.112	1395	0.145	1782	0.178	2169
0.113	1407	0.146	1793	0.179	2180
0.114	1418	0.147	1805	0.180	2192
0.115	1430	0.148	1817	0.181	2204
0.116	1102	0.149	1829	0.182	2215
0.117	1114	0.150	1840	0.183	2227
0.118	1125	0.151	1852	0.184	2239
0.119	1137	0.152	1864	0.185	2251

1/ Specification requirements begin for this ordered thickness

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-II. Minimum required ballistic limits (protection criteria) for Class 2 armor, firing obliquity: 0°, projectile: caliber .30 FSP - Continued.

Thickness, inches	Required BL(P), ft/sec	Thickness, inches	Required BL(P), ft/sec	Thickness, inches	Required BL(P), ft/sec
0.186	2262	0.215	2602	0.244	2942
0.187	2274	0.216	2614	0.245	2954
0.188	2286	0.217	2626	0.246	2966
0.189	2297	0.218	2637	0.247	2977
0.190	2309	0.219	2649	0.248	2989
0.191	2321	0.220	2661	0.249	3001
0.192	2333	0.221	2673	0.250 2/	3013
0.193	2344	0.222	2684	0.251	3024
0.194	2356	0.223	2696	0.252	3036
0.195	2368	0.224	2708	0.253	3048
0.196	2380	0.225	2719	0.254	3059
0.197	2391	0.226	2731	0.255	3071
0.198	2403	0.227	2743	0.256	3083
0.199	2415	0.228	2755	0.257	3095
0.200	2426	0.229	2766	0.258	3106
0.201	2438	0.230	2778	0.259	3118
0.202	2450	0.231	2790	0.260	3130
0.203	2462	0.232	2802	0.261	3141
0.204	2473	0.233	2813	0.262	3153
0.205	2485	0.234	2825	0.263	3165
0.206	2497	0.235	2837	0.264	3177
0.207	2508	0.236	2848	0.265	3188
0.208	2520	0.237	2860	0.266	3200
0.209	2532	0.238	2872	0.267	3212
0.210	2544	0.239	2884	0.268	3223
0.211	2555	0.240	2895	0.269	3235
0.212	2567	0.241	2907	0.270	3247
0.213	2579	0.242	2919	0.271	3259
0.214	2591	0.243	2930	0.272	3270

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-III. Minimum required ballistic limits (protection criteria) for Class 1 & 4 armor, firing obliquity: 0°, projectile: caliber .30 M2 AP.

Thickness, inches	Required BL(P), ft/sec **CLASS 1 & 4**	Thickness, inches	Required BL(P), ft/sec **CLASS 1 & 4**	Thickness, inches	Required BL(P), ft/sec **CLASS 1 & 4**
0.230	1509	0.370	2045	0.515	2484
0.235	1532	0.375	2062	0.520	2498
0.240	1554	0.380	2078	0.525	2512
0.245	1575	0.385	2095	0.530	2525
0.250	1596	0.390	2111	0.535	2539
0.251 1/	1600	0.395	2127	0.540	2552
0.255	1618	0.400	2143	0.545	2566
0.260	1638	0.405	2159	0.550	2579
0.265	1659	0.410	2175	0.555	2593
0.270	1679	0.415	2191	0.560	2606
0.275	1699	0.420	2206	0.565	2619
0.280	1719	0.425	2222	0.570	2632
0.285	1739	0.430	2237	0.575	2645
0.290	1758	0.435	2252	0.580	2658
0.295	1778	0.440	2267	0.585	2671
0.300	1797	0.445	2282	0.590	2684
0.305	1815	0.450	2297	0.595	2697
0.310	1834	0.455	2312	0.600	2710
0.315	1853	0.460	2327	0.605	2723
0.320	1871	0.465	2342	0.610	2736
0.325	1889	0.470	2356	0.615	2749
0.330	1907	0.475	2371	0.620	2762
0.335	1925	0.480	2385	0.624 2/	2772
0.340	1942	0.485	2400	0.625	2775
0.345	1960	0.490	2414	0.630	2788
0.330	1977	0.495	2428	0.635	2801
0.355	1994	0.500	2442	0.640	2814
0.360	2011	0.505	2456	0.645	2827
0.365	2028	0.510	2470	0.650	2840

1/ Specification requirements begin for this ordered thickness

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-IV. Minimum required ballistic limits (protection criteria) for Class 2 armor, firing obliquity: 0°, projectile: caliber .50 FSP.

Thickness, inches	Required BL(P), ft/sec	Thickness, inches	Required BL(P), ft/sec	Thickness, inches	Required BL(P), ft/sec
0.230	1730	0.335	2432	0.445	3168
0.235	1763	0.340	2466	0.450	3202
0.240	1797	0.345	2499	0.455	3235
0.245	1830	0.330	2399	0.460	3268
0.250	1864	0.355	2566	0.465	3302
0.251 1/	1870	0.360	2599	0.470	3335
0.255	1897	0.365	2633	0.475	3369
0.260	1930	0.370	2666	0.480	3402
0.265	1964	0.375	2700	0.485	3436
0.270	1997	0.380	2733	0.490	3469
0.275	2031	0.385	2767	0.495	3503
0.280	2064	0.390	2800	0.500 2/	3536
0.285	2098	0.395	2834	0.505	3569
0.290	2131	0.400	2867	0.510	3603
0.295	2165	0.405	2900	0.515	3636
0.300	2198	0.410	2934	0.520	3670
0.305	2231	0.415	2967	0.525	3703
0.310	2265	0.420	3001	0.530	3737
0.315	2298	0.425	3034	0.535	3770
0.320	2332	0.430	3068	0.540	3804
0.325	2365	0.435	3101	0.545	3837
0.330	2399	0.440	3135	0.550	3871

1/ Specification requirements begin for this ordered thickness

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-V. Minimum required ballistic limits (protection criteria) for Class 1, 3 & 4 armor and the upper ballistic limits for Class 3 armor; firing obliquity: 0°, projectile: caliber .50 M2 AP.

Thickness, inches	BL(P), ft/sec CLASS		Thickness, inches	BL(P), ft/sec CLASS		Thickness, inches	BL(P), ft/sec CLASS	
	1, 3 & 4 Min	3 Max		1, 3 & 4 Min	3 Max		1, 3 & 4 Min	3 Max
0.550	1901	2047	0.695	2182	2328	0.845	2439	2585
0.555	1911	2057	0.700	2191	2337	0.850	2447	2593
0.560	1922	2068	0.705	2200	2346	0.855	2455	2601
0.565	1932	2078	0.710	2209	2355	0.860	2464	2610
0.566	1934	2080	0.715	2218	2364	0.865	2472	2618
0.570	1942	2088	0.720	2227	2373	0.870	2480	2626
0.575	1952	2098	0.725	2235	2381	0.875	2488	2634
0.580	1962	2108	0.730	2244	2390	0.880	2496	2642
0.585	1972	2118	0.735	2253	2399	0.885	2504	2650
0.590	1982	2128	0.740	2262	2408	0.890	2511	2657
0.595	1992	2138	0.745	2271	2417	0.895	2519	2665
0.600	2002	2148	0.750	2279	2425	0.900	2527	2673
0.605	2012	2158	0.755	2288	2434	0.905	2535	2681
0.610	2022	2168	0.760	2297	2443	0.910	2543	2689
0.615	2031	2177	0.765	2305	2451	0.915	2551	2697
0.620	2041	2187	0.770	2314	2460	0.920	2559	2705
0.625 [1/]	2051	2197	0.775	2322	2468	0.925	2566	2712
0.630	2060	2206	0.780	2331	2477	0.930	2574	2720
0.635	2070	2216	0.785	2339	2485	0.935	2582	2728
0.640	2079	2225	0.790	2348	2494	0.940	2590	2736
0.645	2089	2235	0.795	2356	2502	0.945	2597	2743
0.650	2098	2244	0.800	2365	2511	0.950	2605	2751
0.655	2108	2254	0.805	2373	2519	0.955	2612	2758
0.660	2117	2263	0.810	2382	2528	0.960	2620	2766
0.665	2126	2272	0.815	2390	2536	0.965	2628	2774
0.670	2136	2282	0.820	2398	2544	0.970	2635	2781
0.675	2145	2291	0.825	2406	2552	0.975	2643	2789
0.680	2154	2300	0.830	2415	2561	0.980	2650	2796
0.685	2163	2309	0.835	2423	2569	0.985	2658	2804
0.690	2173	2319	0.840	2431	2577	0.990	2665	2811

[1/] Specification requirements begin for this ordered thickness

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-V. Minimum required ballistic limits (protection criteria) for Class 1, 3 & 4 armor and the upper ballistic limits for Class 3 armor; firing obliquity: 0°, projectile: caliber .50 M2 AP- Continued.

Thickness, inches	BL(P), ft/sec CLASS		Thickness, inches	BL(P), ft/sec CLASS		Thickness, inches	BL(P), ft/sec CLASS	
	1, 3 & 4 Min	3 Max		1, 3 & 4 Min	3 Max		1, 3 & 4 Min	3 Max
0.995	2673	2819	1.065	2775	2921	1.135	2874	3020
1.000	2680	2826	1.070	2782	2928	1.140	2881	3027
1.005	2688	2834	1.075	2790	2936	1.145	2888	3034
1.010	2695	2841	1.080	2797	2943	1.150	2895	3041
1.015	2702	2848	1.085	2804	2950	1.155	2902	3048
1.020	2710	2856	1.090	2811	2957	1.160	2909	3055
1.025	2717	2863	1.095	2818	2964	1.165	2915	3061
1.030	2724	2870	1.100	2825	2971	1.170	2922	3068
1.035	2732	2878	1.105	2832	2978	1.175	2929	3075
1.040	2739	2885	1.110	2839	2985	1.180	2936	3082
1.045	2746	2892	1.115	2846	2992	1.185	2943	3089
1.050	2754	2900	1.120	2853	2999	1.190	2949	3095
1.055	2761	2907	1.125 2/	2860	3006	1.195	2956	3102
1.060	2768	2914	1.130	2867	3013	1.200	2963	3109

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-VI. Minimum required ballistic limits (protection criteria) for Class 2 armor, firing obliquity: 0°, projectile: 20 mm FSP.

Thickness, inches	Required BL(P), ft/sec	Thickness, inches	Required BL(P), ft/sec	Thickness, inches	Required BL(P), ft/sec
0.445	1937	0.590	2417	0.740	2914
0.450	1954	0.595	2434	0.745	2931
0.455	1971	0.600	2451	0.750	2947
0.460	1987	0.605	2467	0.755	2964
0.465	2004	0.610	2484	0.760	2980
0.470	2020	0.615	2500	0.765	2997
0.475	2037	0.620	2517	0.770	3013
0.480	2053	0.625 1/	2533	0.775	3030
0.485	2070	0.630	2550	0.780	3047
0.490	2086	0.635	2566	0.785	3063
0.495	2103	0.640	2583	0.790	3080
0.501 1/	2123	0.645	2600	0.795	3096
0.505	2136	0.650	2616	0.800	3113
0.510	2153	0.655	2633	0.805	3129
0.515	2169	0.660	2649	0.810	3146
0.520	2186	0.665	2666	0.815	3162
0.525	2202	0.670	2682	0.820	3179
0.530	2219	0.675	2699	0.825	3196
0.535	2235	0.680	2715	0.830	3212
0.540	2252	0.685	2732	0.835	3229
0.545	2268	0.690	2749	0.840	3245
0.550	2285	0.695	2765	0.845	3262
0.555	2302	0.700	2782	0.850	3278
0.560	2318	0.705	2798	0.855	3295
0.565	2335	0.710	2815	0.860	3311
0.566	2338	0.715	2831	0.865	3328
0.570	2351	0.720	2848	0.870	3345
0.575	2368	0.725	2864	0.875	3361
0.580	2384	0.730	2881	0.880	3378
0.585	2401	0.735	2898	0.885	3394

1/ Specification requirements begin for this ordered thickness

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

TABLE A-VI. Minimum required ballistic limits (protection criteria) for Class 2 armor, firing obliquity: 0°, projectile: 20 mm FSP - Continued.

Thickness, inches	Required BL(P), ft/sec	Thickness, inches	Required BL(P), ft/sec	Thickness, inches	Required BL(P), ft/sec
0.890	3411	0.945	3593	1.000 [2]	3775
0.895	3427	0.950	3609	1.005	3792
0.900	3444	0.955	3626	1.010	3808
0.905	3460	0.960	3643	1.015	3825
0.910	3477	0.965	3659	1.020	3841
0.915	3494	0.970	3676	1.025	3858
0.920	3510	0.975	3692	1.030	3874
0.925	3527	0.980	3709	1.035	3891
0.930	3543	0.985	3725	1.040	3907
0.935	3560	0.990	3742	1.045	3924
0.940	3576	0.995	3758	1.050	3941

[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-VII. Minimum required ballistic limits (protection criteria) for Class 1, 3 & 4 armor and the upper ballistic limits for Class 3 armor; firing obliquity: 0°, projectile: 20 mm M602 APIT.

Thickness, inches	BL(P), ft/sec CLASS		Thickness, inches	BL(P), ft/sec CLASS		Thickness, inches	BL(P), ft/sec CLASS	
	1, 3 & 4 Min	3 Max		1, 3 & 4 Min	3 Max		1, 3 & 4 Min	3 Max
1.080	1909	2016	1.400	2265	2403	1.730	2586	2748
1.090	1921	2029	1.410	2275	2414	1.740	2595	2758
1.100	1933	2042	1.420	2285	2425	1.750	2604	2768
1.110	1945	2055	1.430	2295	2436	1.760	2613	2778
1.120	1957	2068	1.440	2305	2447	1.770	2613	2787
1.126 [1/]	1964	2076	1.450	2315	2458	1.780	2621	2797
1.130	1969	2081	1.460	2325	2469	1.790	2630	2807
1.140	1981	2094	1.470	2335	2480	1.800	2639	2816
1.150	1993	2106	1.480	2344	2491	1.810	2647	2826
1.160	2004	2119	1.490	2354	2501	1.820	2656	2836
1.170	2016	2131	1.500	2364	2512	1.830	2664	2845
1.180	2027	2144	1.510	2374	2523	1.840	2673	2855
1.190	2039	2156	1.520	2383	2534	1.850	2681	2864
1.200	2050	2169	1.530	2393	2544	1.860	2690	2874
1.210	2061	2181	1.540	2403	2555	1.870	2698	2883
1.220	2073	2193	1.550	2412	2565	1.880	2707	2892
1.230	2084	2205	1.560	2422	2576	1.890	2715	2902
1.240	2095	2217	1.570	2431	2586	1.900	2723	2911
1.250	2106	2229	1.580	2440	2597	1.910	2732	2921
1.260	2117	2241	1.590	2450	2607	1.920	2740	2930
1.270	2128	2253	1.600	2459	2617	1.930	2748	2939
1.280	2139	2265	1.610	2468	2628	1.940	2757	2948
1.290	2149	2277	1.620	2478	2638	1.950	2765	2958
1.300	2160	2289	1.630	2487	2648	1.960	2773	2967
1.310	2171	2300	1.640	2496	2658	1.970	2781	2976
1.320	2181	2312	1.650	2505	2668	1.980	2789	2985
1.330	2192	2323	1.660	2514	2678	1.990	2798	2994
1.340	2203	2335	1.670	2524	2689	2.000 [2/]	2806	3003
1.350	2213	2346	1.680	2533	2699	2.010	2814	3012
1.360	2223	2358	1.690	2542	2709	2.020	2822	3021
1.370	2234	2369	1.700	2551	2719	2.030	2830	3030
1.380	2244	2380	1.710	2560	2728	2.040	3838	3039
1.390	2254	2392	1.720	2568	2738	2.050	3846	3048

[1/] Specification requirements begin for this ordered thickness.

[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-VIII. Minimum required ballistic limits (protection criteria) for Class 1 and 3 armor and the upper ballistic limits for Class 3 armor; firing obliquity: 0°, projectile: 20 mm M602 APIT.

Thickness, inches	BL(P), ft/sec CLASS		Thickness, inches	BL(P), ft/sec CLASS		Thickness, inches	BL(P), ft/sec CLASS	
	1 & 3 Min	3 Max		1 & 3 Min	3 Max		1 & 3 Min	3 Max
1.930	2748	2939	2.240	2993	3214	2.550	3217	3469
1.940	2757	2948	2.250	3000	3223	2.560	3224	3477
1.950	2765	2958	2.260	3008	3231	2.570	3231	3485
1.960	2773	2967	2.270	3015	3240	2.580	3238	3493
1.970	2781	2976	2.280	3023	3248	2.590	3245	3501
1.980	2789	2985	2.290	3030	3257	2.600	3252	3509
1.990	2798	2994	2.300	3038	3265	2.610	3259	3516
2.001 [1/]	2807	3004	2.310	3045	3273	2.620	3265	3524
2.010	2814	3012	2.320	3052	3282	2.630	3272	3532
2.020	2822	3021	2.330	3060	3290	2.640	3279	3540
2.030	2830	3030	2.340	3067	3298	2.650	3286	3548
2.040	2838	3039	2.350	3075	3307	2.660	3293	3556
2.050	2846	3048	2.360	3082	3315	2.670	3299	3563
2.060	2854	3057	2.370	3089	3323	2.680	3306	3571
2.070	2862	3066	2.380	3096	3331	2.690	3313	3579
2.080	2870	3075	2.390	3104	3340	2.700	3320	3586
2.090	2877	3084	2.400	3111	3348	2.710	3326	3594
2.100	2885	3093	2.410	3118	3356	2.720	3333	3602
2.110	2893	3102	2.420	3125	3364	2.730	3340	3610
2.120	2901	3110	2.430	3132	3372	2.740	3346	3617
2.130	2909	3119	2.440	3140	3381	2.750 [2/]	3353	3625
2.140	2916	3128	2.450	3147	3389	2.760	3360	3632
2.150	2924	3137	2.460	3154	3397	2.770	3366	3640
2.160	2932	3145	2.470	3161	3405	2.780	3373	3648
2.170	2940	3154	2.480	3168	3413	2.790	3380	3655
2.180	2947	3163	2.490	3175	3421	2.800	3386	3663
2.190	2955	3171	2.500	3182	3429	2.810	3393	3670
2.200	2963	3180	2.510	3189	3437	2.820	3400	3678
2.210	2970	3188	2.520	3196	3445	2.830	3406	3686
2.220	2978	3197	2.530	3203	3453	2.840	3413	3694
2.230	2985	3206	2.540	3210	3461	2.850	3420	3701

[1/] Specification requirements begin for this ordered thickness.
[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-IX. Minimum required ballistic limits (protection criteria) for Class 1 and 3 armor and the upper ballistic limits for Class 3 armor; firing obliquity: 45°, projectile: caliber 90 mm M82 APC.

Thickness, inches	BL(P), ft/sec CLASS		Thickness, inches	BL(P), ft/sec CLASS		Thickness, inches	BL(P), ft/sec CLASS	
	1 & 3 Min	3 Max		1 & 3 Min	3 Max		1 & 3 Min	3 Max
2.690	2089	2382	2.980	2260	2553	3.280	2425	2718
2.700	2095	2388	2.990	2265	2558	3.290	2430	2723
2.710	2101	2394	3.000	2271	2564	3.300	2436	2729
2.720	2107	2400	3.010	2277	2570	3.310	2441	2734
2.730	2113	2406	3.020	2282	2575	3.320	2446	2739
2.740	2119	2412	3.030	2288	2581	3.330	2451	2744
2.750	2125	2418	3.040	2294	2587	3.340	2457	2750
2.751 [1/]	2126	2419	3.050	2299	2592	3.350	2462	2755
2.760	2131	2424	3.060	2305	2598	3.360	2467	2760
2.770	2137	2430	3.070	2310	2603	3.370	2472	2765
2.780	2143	2436	3.080	2316	2609	3.380	2478	2771
2.790	2149	2442	3.090	2322	2615	3.390	2483	2776
2.800	2155	2448	3.100	2327	2620	3.400	2488	2781
2.810	2161	2454	3.110	2333	2626	3.410	2493	2786
2.820	2167	2460	3.120	2338	2631	3.420	2498	2791
2.830	2173	2466	3.130	2344	2637	3.430	2404	2697
2.840	2179	2472	3.140	2349	2642	3.440	2509	2802
2.850	2185	2478	3.150	2355	2648	3.450	2514	2807
2.860	2190	2483	3.160	2360	2653	3.460	2519	2812
2.870	2196	2489	3.170	2366	2659	3.470	2524	2817
2.880	2202	2495	3.180	2371	2664	3.480	2529	2822
2.890	2208	2501	3.190	2376	2669	3.490	2534	2827
2.900	2214	2507	3.200	2382	2675	3.500 [2/]	2540	2833
2.910	2220	2513	3.210	2387	2680	3.510	2545	2838
2.920	2225	2518	3.220	2393	2686	3.520	2550	2843
2.930	2231	2524	3.230	2398	2691	3.530	2555	2848
2.940	2237	2530	3.240	2404	2697	3.540	2560	2853
2.950	2243	2536	3.250	2409	2702	3.550	2565	2858
2.960	2248	2541	3.260	2414	2707	3.560	2570	2863
2.970	2254	2547	3.270	2420	2713	3.570	2575	2868

[1/] Specification requirements begin for this ordered thickness.

[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-X. Minimum required ballistic limits (protection criteria) for Class 1 and 3 armor and the upper ballistic limits for Class 3 armor; firing obliquity: 30°, projectile: 90 mm M82 APC.

Thickness, inches	BL(P), ft/sec CLASS		Thickness, inches	BL(P), ft/sec CLASS		Thickness, inches	BL(P), ft/sec CLASS	
	1 & 3 Min	3 Max		1 & 3 Min	3 Max		1 & 3 Min	3 Max
3.425	1924	2116	3.570	1986	2178	3.720	2048	2240
3.430	1926	2118	3.575	1988	2180	3.725	2051	2243
3.435	1929	2121	3.580	1990	2182	3.730	2053	2245
3.440	1931	2123	3.585	1993	2185	3.735	2055	2247
3.445	1933	2125	3.590	1995	2187	3.740	2057	2249
3.450	1935	2127	3.595	1997	2189	3.745	2069	2261
3.455	1937	2129	3.600	1999	2191	3.750	2061	2253
3.460	1939	2131	3.605	2001	2193	3.755	2063	2255
3.465	1942	2134	3.610	2003	2195	3.760	2065	2257
3.470	1944	2136	3.615	2005	2197	3.765	2067	2259
3.475	1946	2138	3.620	2007	2199	3.770	2069	2261
3.480	1948	2140	3.625	2009	2201	3.775	2071	2263
3.485	1950	2142	3.630	2011	2203	3.780	2073	2265
3.490	1952	2144	3.635	2013	2205	3.785	2075	2267
3.495	1954	2146	3.640	2015	2207	3.790	2077	2269
3.500	1957	2149	3.645	2018	2210	3.795	2079	2271
3.501 1/	1957	2149	3.650	2020	2212	3.800	2081	2273
3.505	1959	2151	3.655	2022	2214	3.805	2083	2275
3.510	1961	2153	3.660	2024	2216	3.810	2085	2277
3.515	1963	2155	3.665	2026	2218	3.815	2087	2279
3.520	1965	2157	3.670	2028	2220	3.820	2089	2281
3.525	1967	2159	3.675	2030	2222	3.825	2091	2283
3.530	1969	2161	3.680	2032	2224	3.830	2093	2285
3.535	1971	2163	3.685	2034	2226	3.835	2095	2287
3.540	1974	2166	3.690	2036	2228	3.840	2097	2289
3.545	1976	2168	3.695	2038	2230	3.845	2099	2291
3.550	1978	2170	3.700	2040	2232	3.850	2101	2293
3.555	1980	2172	3.705	2042	2234	3.855	2103	2295
3.560	1982	2174	3.710	2044	2236	3.860	2105	2297
3.565	1984	2176	3.715	2046	2238	3.865	2107	2299

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

TABLE A-X. Minimum required ballistic limits (protection criteria) for Class 1 and 3 armor and the upper ballistic limits for Class 3 armor; firing obliquity: 30°,projectile: 90 mm M82 APC – Continued.

Thickness, inches	BL(P), ft/sec CLASS		Thickness, inches	BL(P), ft/sec CLASS		Thickness, inches	BL(P), ft/sec CLASS	
	1 & 3 Min	3 Max		1 & 3 Min	3 Max		1 & 3 Min	3 Max
3.870	2109	2301	3.940	2137	2329	4.005	2162	2354
3.875	2111	2303	3.945	2139	2331	4.010	2164	2356
3.880	2113	2305	3.950	2141	2333	4.015	2166	2358
3.885	2115	2307	3.955	2143	2335	4.020	2168	2360
3.890	2117	2309	3.960	2144	2336	4.025	2170	2362
3.895	2119	2311	3.965	2146	2338	4.030	2172	2364
3.900	2121	2313	3.970	2148	2340	4.035	2174	2366
3.905	2123	2315	3.975	2150	2342	4.040	2176	2368
3.910	2125	2317	3.980	2152	2344	4.045	2178	2370
3.915	2127	2319	3.985	2154	2346	4.050	2179	2371
3.920	2129	2321	3.990	2156	2348	4.055	2181	2373
3.925	2131	2323	3.995	2158	2350	4.060	2183	2375
3.930	2133	2325	3.999 [2/]	2160	2352	4.065	2185	2377
3.935	2135	2327	4.000	2160	2352	4.070	2187	2379

[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

TABLE A-XI. Minimum required ballistic limits (protection criteria) for Class 1 and 3 armor and the upper ballistic limits for Class 3 armor; firing obliquity: 0°, projectile: 90 mm M318A1 AP.

Thickness, inches	BL(P), ft/sec CLASS		Thickness, inches	BL(P), ft/sec CLASS		Thickness, inches	BL(P), ft/sec CLASS	
	1 & 3 Min	3 Max		1 & 3 Min	3 Max		1 & 3 Min	3 Max
3.940	1583	1785	4.260	1697	1899	4.580	1805	2007
3.950	1586	1788	4.270	1701	1903	4.590	1808	2010
3.960	1590	1792	4.280	1704	1906	4.600	1811	2013
3.970	1594	1796	4.290	1708	1910	4.610	1815	2017
3.980	1597	1799	4.300	1711	1913	4.620	1818	2020
3.990	1601	1803	4.310	1714	1916	4.630	1821	2023
4.000 [1/]	1605	1807	4.320	1718	1920	4.640	1824	2026
4.010	1608	1810	4.330	1721	1923	4.650	1827	2029
4.020	1612	1814	4.340	1725	1927	4.660	1831	2033
4.030	1616	1818	4.350	1728	1930	4.670	1834	2036
4.040	1619	1821	4.360	1731	1933	4.680	1837	2039
4.050	1623	1825	4.370	1735	1937	4.690	1840	2042
4.060	1627	1829	4.380	1738	1940	4.700	1844	2046
4.070	1630	1832	4.390	1742	1944	4.710	1847	2049
4.080	1634	1836	4.400	1745	1947	4.720	1850	2052
4.090	1637	1839	4.410	1748	1950	4.730	1853	2055
4.100	1641	1843	4.420	1752	1954	4.740	1856	2058
4.110	1644	1846	4.430	1755	1957	4.750	1860	2062
4.120	1648	1850	4.440	1758	1960	4.760	1863	2065
4.130	1652	1854	4.450	1762	1964	4.770	1866	2068
4.140	1655	1857	4.460	1765	1967	4.780	1869	2071
4.150	1659	1861	4.470	1768	1970	4.790	1872	2074
4.160	1662	1864	4.480	1772	1974	4.800	1875	2077
4.170	1666	1868	4.490	1775	1977	4.810	1878	2080
4.180	1669	1871	4.500	1778	1980	4.820	1882	2084
4.190	1673	1875	4.510	1782	1984	4.830	1885	2087
4.200	1676	1878	4.520	1788	1990	4.840	1888	2090
4.210	1680	1882	4.530	1792	1994	4.850	1891	2093
4.220	1683	1885	4.540	1795	1997	4.860	1894	2096
4.230	1687	1889	4.550	1798	2000	4.870	1897	2099
4.240	1690	1892	4.560	1783	1985	4.880	1900	2102
4.250	1694	1896	4.570	1801	2003	4.890	1903	2105

[1/] Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

APPENDIX A

TABLE A-XI. Minimum required ballistic limits (protection criteria) for Class 1 and 3 armor and the upper ballistic limits for Class 3 armor; firing obliquity: 0°,projectile: 90 mm M318A1 AP – Continued.

Thickness, inches	BL(P), ft/sec CLASS		Thickness, inches	BL(P), ft/sec CLASS		Thickness, inches	BL(P), ft/sec CLASS	
	1 & 3 Min	3 Max		1 & 3 Min	3 Max		1 & 3 Min	3 Max
4.900	1907	2109	5.230	2006	2208	5.560	2102	2304
4.910	1910	2112	5.240	2009	2211	5.570	2105	2307
4.920	1913	2115	5.250	2012	2214	5.580	2107	2309
4.930	1916	2118	5.260	2015	2217	5.590	2110	2312
4.940	1919	2121	5.270	2018	2220	5.600	2113	2315
4.950	1922	2124	5.280	2021	2223	5.610	2116	2318
4.960	1925	2127	5.290	2024	2226	5.620	2119	2321
4.970	1928	2130	5.300	2027	2229	5.630	2121	2323
4.980	1931	2133	5.310	2030	2232	5.640	2124	2326
4.990	1934	2136	5.320	2033	2235	5.650	2127	2329
5.000	1937	2139	5.330	2036	2238	5.660	2130	2332
5.010	1940	2142	5.340	2039	2241	5.670	2133	2335
5.020	1943	2145	5.350	2042	2244	5.680	2135	2337
5.030	1946	2148	5.360	2045	2247	5.690	2138	2340
5.040	1950	2152	5.370	2047	2249	5.700	2141	2343
5.050	1953	2155	5.380	2050	2252	5.710	2144	2346
5.060	1956	2158	5.390	2053	2255	5.720	2147	2349
5.070	1959	2161	5.400	2056	2258	5.730	2149	2351
5.080	1962	2164	5.410	2059	2261	5.740	2152	2354
5.090	1965	2167	5.420	2062	2264	5.750	2155	2357
5.110	1968	2170	5.430	2065	2267	5.760	2158	2360
5.110	1971	2173	5.440	2068	2270	5.770	2160	2362
5.120	1974	2176	5.450	2070	2272	5.780	2163	2365
5.130	1977	2179	5.460	2073	2275	5.790	2166	2368
5.140	1980	2182	5.470	2076	2278	5.800	2169	2371
5.150	1983	2185	5.480	2079	2281	5.810	2171	2373
5.160	1986	2188	5.490	2082	2284	5.820	2174	2376
5.170	1989	2191	5.500	2085	2287	5.830	2177	2379
5.180	1992	2194	5.510	2088	2290	5.840	2180	2382
5.190	1995	2197	5.520	2090	2292	5.850	2182	2384
5.200	1998	2200	5.530	2093	2295	5.860	2185	2387
5.210	2001	2203	5.540	2096	2298	5.870	2188	2390
5.220	2003	2205	5.550	2099	2301	5.880	2190	2392

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

TABLE A-XI. Minimum required ballistic limits (protection criteria) for Class 1 and 3 armor and the upper ballistic limits for Class 3 armor; firing obliquity: 0°, projectile: 90 mm M318A1 AP – Continued.

Thickness, inches	BL(P), ft/sec		Thickness, inches	BL(P), ft/sec		Thickness, inches	BL(P), ft/sec	
	C LASS			CLASS			CLASS	
	1 & 3 Min	3 Max		1 & 3 Min	3 Max		1 & 3 Min	3 Max
5.890	2193	2395	5.950	2209	2411	6.010	2226	2428
5.900	2196	2398	5.960	2212	2414	6.020	2228	2430
5.910	2199	2401	5.970	2215	2417	6.030	2231	2433
5.920	2201	2403	5.980	2217	2419	6.040	2234	2436
5.930	2204	2406	5.990	2220	2422	6.050	2236	2438
5.940	2207	2409	6.000 [2/]	2223	2425	6.060	2239	2441

[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the specification requirements are for interpolation of BL(P) requirements on undersize or oversize plates.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

APPENDIX A

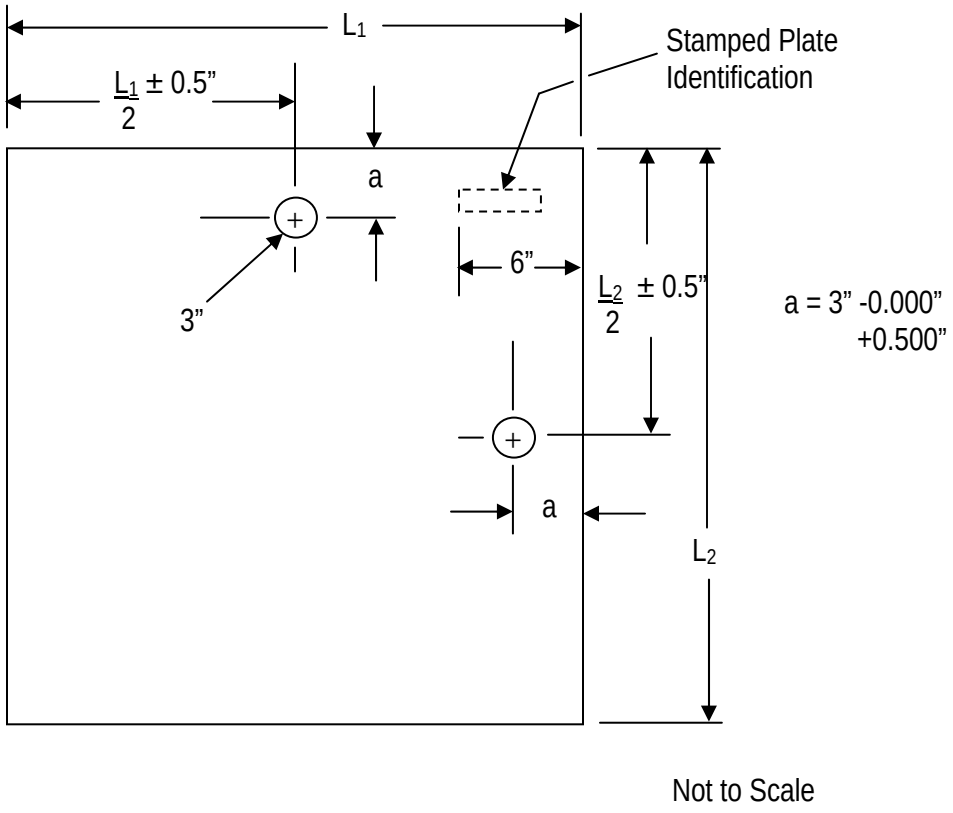


FIGURE A-1. Lifting holes for ballistic acceptance test plates 2.751 through 6.25 inches thick.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodian:
Army - MR

Preparing activity:
Army - MR

Review activities:
Army - AR, AT, AV, TE
DLA - IS

Project 9515-2007-008

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T23:00Z
Check the source to verify that this is the current version before use.